

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401160
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Yang; Yuchen et al.

Connector and connector assembly

Abstract

A connector includes a housing having a mounting port, a connection assembly installed in the mounting port of the housing and adapted to fasten a terminal of the connector to a mating terminal of a mating connector, the connection assembly has a first operation part exposed from the housing and the connection assembly is capable of being fastened or loosened by operating the first operation part, and a safety cover slidably mounted on the housing. The first operation part is in a position of interference with the safety cover to prevent the safety cover from being moved to a safety position when the connection assembly is not fastened.

Inventors: Yang; Yuchen (Shanghai, CN), Zhou; Xiao (Nichee) (Shanghai, CN), Jiang; Kaixuan (Evan) (Shanghai, CN), Li; Ziwei (Vivi) (Shanghai, CN)

Applicant: Tyco Electronics (Shanghai) Co., Ltd. (Shanghai, CN)

Family ID: 1000008781417

Assignee: Tyco Electronics (Shanghai) Co., Ltd. (Shanghai, CN)

Appl. No.: 18/181188

Filed: March 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230307864 A1	Sep. 28, 2023

Foreign Application Priority Data

CN	202210225027.5	Mar. 09, 2022
CN	202220496338.0	Mar. 09, 2022

Publication Classification

Int. Cl.: H01R13/641 (20060101); H01R13/621 (20060101)

U.S. Cl.:

CPC H01R13/641 (20130101); H01R13/6215 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6419522	12/2001	Bonilla	439/468	H01R 13/506
7270562	12/2006	Trout	439/364	H01R 13/6215
7524215	12/2008	Banas	439/465	H01R 13/506
7625229	12/2008	Cross	439/468	H01R 13/5202
8052481	12/2010	Azad	439/695	H01R 13/6215
8177576	12/2011	Zhou	439/372	H01R 13/6215
9059540	12/2014	Menez	N/A	H01R 13/641
9105998	12/2014	Uno	N/A	H01R 13/6315
10211570	12/2018	Tanaka	N/A	H01R 13/6215
11223157	12/2021	Nakano	N/A	H01R 13/5219
11322880	12/2021	Tanaka	N/A	H01R 13/5202
11699885	12/2022	Pavlovic	235/451	H01R 13/6581
11942729	12/2023	Yamada	N/A	H01R 13/533
11955751	12/2023	Nishijima	N/A	H01R 13/6592
12046858	12/2023	Martin	N/A	H01R 13/641
12244096	12/2024	Martin	N/A	H01R 13/6271
2014/0120763	12/2013	Itsuki	439/382	H01R 13/533
2017/0256893	12/2016	Kato	N/A	H01R 13/6593
2023/0123353	12/2022	Qian	439/490	H01R 13/641
2023/0187878	12/2022	Qian	439/489	H01R 13/701
2023/0291150	12/2022	Yang	N/A	H01R 13/701
2024/0063574	12/2023	Simbürger	N/A	H01R 13/6273
2024/0106175	12/2023	Li	N/A	H01R 13/6585
2024/0266777	12/2023	Matsuoka	N/A	H01R 13/504
2024/0317083	12/2023	Yamaguchi	N/A	H01R 13/04
2024/0413577	12/2023	Wimmer	N/A	H01R 13/465
2025/0023293	12/2024	Guillanton	N/A	H01R 13/748
2025/0030192	12/2024	Fang	N/A	H01R 13/665

Primary Examiner: Gushi; Ross N

Attorney, Agent or Firm: Barley Snyder

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of the filing date under 35 U.S.C. § 119 (a)-(d) of Chinese Patent Application No. 202210225027.5, filed on Mar. 9, 2022, and Chinese Patent Application No. 202220496338.0, filed on Mar. 9, 2022.

FIELD OF THE INVENTION

(2) The present invention relates to a connector and a connector assembly including the connector.

BACKGROUND

(3) In the prior art, the connector at the cable end usually includes a housing, a terminal, a connection assembly and a cable. The terminal is arranged in the housing, and the cable extends into the housing and is electrically connected with the terminal. The connection assembly is installed in the housing, which is used to fasten the terminal to a mating terminal of a mating connector, so that the terminal is electrically connected with the mating terminal.

(4) In the prior art, when the connector is mated with the mating connector, there is a lack of safety protection device to judge whether the connection assembly has been fastened to prevent the fastened connection assembly from being accidentally loosened, which reduces the use safety of the connector.

(5) In addition, in the prior art, in order to detect the connection state between the terminal of the connector and the mating terminal of the mating connector, it is also necessary to set a high-voltage interlock (or called a detection connector) and a mating high-voltage interlock (or called a mating detection connector) on the connector and the mating connector respectively. However, the existing high-voltage interlock is fixed on the housing of the connector, so there is a risk that the high-voltage interlock will be mated with the mating high-voltage interlock before the terminal of the connector is electrically connected with the mating terminal of the mating connector, which also reduces the use safety of the connector.

SUMMARY

(6) A connector includes a housing having a mounting port, a connection assembly installed in the mounting port of the housing and adapted to fasten a terminal of the connector to a mating terminal of a mating connector, the connection assembly has a first operation part exposed from the housing and the connection assembly is capable of being fastened or loosened by operating the first operation part, and a safety cover slidably mounted on the housing. The first operation part is in a position of interference with the safety cover to prevent the safety cover from being moved to a safety position when the connection assembly is not fastened.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of the present invention will become more apparent by describing in detail exemplary embodiments thereof with reference to the accompanying drawings, in which:

(2) FIG. 1 shows an illustrative exploded view of a connector assembly according to an exemplary embodiment of the present invention, in which the safety cover of the connector is not shown;

(3) FIG. 2 shows an illustrative exploded view of the connector assembly according to an exemplary embodiment of the present invention, in which the safety cover and housing of the connector are not shown;

(4) FIG. 3 shows an illustrative assembly view of the connector assembly according to an exemplary embodiment of the present invention, wherein the safety cover and housing of the connector are not shown;

(5) FIG. 4 shows an illustrative view of a connector assembly according to an exemplary embodiment of the present invention, wherein the connection assembly and the mechanical

connection member are in the loosened position, and the safety cover is in the initial position;

(6) FIG. 5 shows an illustrative view of the connector assembly according to an exemplary embodiment of the present invention, wherein the connection assembly and the mechanical connection member are in the fastened position, and the safety cover is in the initial position;

(7) FIG. 6 shows an illustrative view of the connector assembly according to an exemplary embodiment of the present invention, wherein the connection assembly and the mechanical connection member are in the fastened position, and the safety cover is moved to the safe position;

(8) FIG. 7 shows an illustrative view of a connector housing according to an exemplary embodiment of the present invention;

(9) FIG. 8 shows an illustrative perspective view of the safety cover of the connector according to an exemplary embodiment of the present invention when viewed from the top;

(10) FIG. 9 shows an illustrative perspective view of the safety cover of the connector according to an exemplary embodiment of the present invention when viewed from the bottom;

(11) FIG. 10 shows a transverse sectional view of a connector housing and a safety cover according to an exemplary embodiment of the present invention;

(12) FIG. 11 shows an illustrative view of a connector according to an exemplary embodiment of the present invention, wherein the connection assembly and mechanical connection member are in the fastened position, and the safety cover is in the initial position;

(13) FIG. 12 shows an illustrative perspective view of a mating connector according to an exemplary embodiment of the present invention;

(14) FIG. 13 shows an illustrative assembly view of the connector housing and the high-voltage interlock according to an exemplary embodiment of the present invention;

(15) FIG. 14 shows an illustrative perspective view of a connector housing according to an exemplary embodiment of the present invention;

(16) FIG. 15 shows an illustrative perspective view of a high-voltage interlock of a connector according to an exemplary embodiment of the present invention;

(17) FIG. 16 shows an illustrative view of a high-voltage interlock and safety cover of a connector and a mating high-voltage interlock of a mating connector according to an exemplary embodiment of the present invention, wherein the safety cover is in the initial position and the high-voltage interlock is in the disconnected position;

(18) FIG. 17 shows an illustrative assembly view of the high-voltage interlock and the safety cover of the connector according to an exemplary embodiment of the present invention, wherein the safety cover is in the initial position;

(19) FIG. 18 shows an illustrative view of the safety cover of the connector according to an exemplary embodiment of the present invention when viewed from the bottom, in which the structure connected with the high-voltage interlock is shown;

(20) FIG. 19 shows the sectional view of the high-voltage interlock and safety cover of the connector and the mating high-voltage interlock of the mating connector according to an exemplary embodiment of the present invention, wherein the safety cover is in the safe position and the high-voltage interlock is in the connected position;

(21) FIG. 20 shows an illustrative view of the high-voltage interlock and safety cover of the connector and the mating high-voltage interlock of the mating connector according to an exemplary embodiment of the present invention, wherein the safety cover is in the safe position and the high-voltage interlock is in the connected position; and

(22) FIG. 21 shows the sectional view of the high-voltage interlock and safety cover of the connector and the mating high-voltage interlock of the mating connector according to an exemplary embodiment of the present invention, wherein the safety cover is in the safe position and the high-voltage interlock is in the connected position.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(23) Exemplary embodiments of the present disclosure will be described hereinafter in detail with

reference to the attached drawings, wherein like reference numerals refer to like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the disclosure to those skilled in the art.

(24) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

(25) As shown in FIGS. **1** to **3**, in the illustrated embodiment, the connector assembly mainly includes a connector **1** and a mating connector **2**. The connector **1** includes a housing **10**, a terminal **13** and a connection assembly **11**. The mating connector **2** includes a mating housing **20** mated with the housing **10** and a mating terminal **21** electrically connected with the terminal **13**.

(26) As shown in FIGS. **1** to **3**, in the illustrated embodiment, terminal **13** is arranged in housing **10**. The housing **10** has a mounting port **1011**. The connection assembly **11** is installed in the mounting port **1011** of the housing **10**, and is suitable for fastening the terminal **13** to the mating terminal **21** of the mating connector **2**. The connection assembly **11** has a first operation part **111** exposed from the housing **10**, which enables the connection assembly **11** to be fastened or loosened by the first operation part **111**.

(27) As shown in FIGS. **1** to **6**, in the illustrated embodiment, the connector **1** also includes a safety cover **100**, which is slidably mounted on the housing **10**.

(28) As shown in FIG. **4**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when the connection assembly **11** is not fastened, the first operation part **111** is in the position of interference with the safety cover **100** (for example, as shown in FIG. **4**, the top surface of the first operation part **111** is higher than the inner surface of the safety cover **100**), to prevent the safety cover **100** from being moved from the initial position to the safe position.

(29) As shown in FIGS. **4** and **5**, in the illustrated embodiment, when the safety cover **100** is in the initial position, the first operation part **111** is not covered by the safety cover **100** to allow the connection assembly **11** to be fastened or loosened by operating the first operation part **111**.

(30) Please note that, as shown in FIGS. **4** and **5**, in an exemplary embodiment of the present invention, when the safety cover **100** is in the initial position, the connection assembly **11** can be fastened (as shown in FIG. **5**) or not be fastened (as shown in FIG. **4**).

(31) As shown in FIG. **6**, in the illustrated embodiment, when the safety cover **100** is in the safe position, the first operation part **111** is covered by the safety cover **100** to prevent the fastened connection assembly **11** from being loosened by operating the first operation part **111**.

(32) As shown in FIGS. **1** to **6**, in the illustrated embodiment, the connector **1** also includes a mechanical connection member **12**, which is installed on the housing **10** and is suitable for threaded connection with the mating housing **20** of the mating connector **2** to fasten the housing **10** onto the mating housing **20**.

(33) As shown in FIGS. **1** to **6**, in the illustrated embodiment, the mechanical connection member **12** has a second operation part **121** exposed from the housing **10**, so that the mechanical connection member **12** is capable of being fastened or loosened by operating the second operation part **121**.

(34) As shown in FIG. **4**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when the mechanical connection member **12** is not fastened, the second operation part **121** is in the position of interference with the safety cover **100** (for example, as shown in FIG. **4**, the top surface of the second operation part **121** is higher than the inner surface of the safety cover **100**) to prevent the safety cover **100** from being moved from the initial position to the safe position.

(35) As shown in FIG. **4** and FIG. **5**, in the illustrated embodiment, when the safety cover **100** is in

the initial position, the second operation part **121** is not covered by the safety cover **100** to allow the mechanical connection member **12** to be fastened or loosened by operating the second operation part **121**.

(36) Please note that, as shown in FIGS. **4** and **5**, in an exemplary embodiment of the present invention, when the safety cover **100** is in the initial position, the mechanical connection member **12** can be fastened (as shown in FIG. **5**) or not be fastened (as shown in FIG. **4**).

(37) As shown in FIG. **6**, in the illustrated embodiment, when the safety cover **100** is in the safe position, the second operation part **121** is covered by the safety cover **100** to prevent the mechanical connection member **12** from being loosened by operating the second operation part **121**.

(38) As shown in FIG. **4**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when either of the connection assembly **11** and the mechanical connection member **12** is not fastened, one of the first operation part **111** and the second operation part **121** will be in the position of interference with the safety cover **100**, so that the safety cover **100** cannot be moved from the initial position to the safe position.

(39) As shown in FIG. **5** and FIG. **6**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when the connection assembly **11** and the mechanical connection member **12** are fastened, the first operation part **111** and the second operation part **121** are in a position that does not interfere with the safety cover **100**. At this time, the safety cover **100** can be moved from the initial position to the safe position.

(40) As shown in FIGS. **1** to **10**, in the illustrated embodiment, the safety cover **100** comprises a top wall **101** and a pair of side walls **102**. An operation hole **101a** is formed on the top wall **101**. A pair of side walls **102** are respectively connected to the top wall **101**. When the safety cover **100** is in the initial position, the first operation part **111** is located outside the safety cover **100**, and the second operation part **121** is exposed from the safety cover **100** through the operation hole **101a**.

(41) As shown in FIGS. **1** to **10**, in the illustrated embodiment, a guide slot **10b** extending in a first direction is formed on one of the outer wall surface of the housing **10** and the inner wall surface of the side wall **102** of the safety cover **100**, and a guide rib **102a** extending in the first direction is formed on the other. The guide rib **102a** is buckled into the guide slot **10b** and can be moved along the guide slot **10b** in the first direction to guide the safety cover **100** to move between the initial position and the safety position relative to the housing **10**.

(42) As shown in FIGS. **1** to **10**, in the illustrated embodiment, the guide slot **10b** is formed on the outer wall surface of the housing **10**, and the guide rib **102a** is formed on the inner wall surface of the side wall **102** of the safety cover **100**. The guide slot **10b** has a guide surface **10c** extending in the first direction, and the guide rib **102a** slides in contact with the guide surface **10c** to guide the guide rib **102a** to move in the first direction.

(43) As shown in FIGS. **1** to **10**, in the illustrated embodiment, a limit part **101b** is formed on the inner wall surface of the top wall **101** of the safety cover **100**. When the safety cover **100** is moved from the initial position to the safe position, the limit part **101b** rests against the first operation part **111** to limit the safety cover **100** at the safe position.

(44) As shown in FIGS. **1** to **10**, in the illustrated embodiment, connector **1** also includes a cable **14** and an end cover **15**. One end of the cable **14** is inserted into the housing **10** in the first direction and electrically connected with the terminal **13**. The end cover **15** is sleeved on one end of the housing **10** to fix the cable **14** to the housing **10**.

(45) As shown in FIGS. **1** to **10**, in the illustrated embodiment, a cable port **1014** allowing the cable **14** to be inserted in the first direction is formed on one end of the housing **10**. The mounting port **1011** is formed on the top of the other end of the housing **10** to allow the connection assembly **11** to be inserted in the second direction perpendicular to the first direction. A mating port is formed at the bottom of the other end of the housing **10** to allow the connector **1** to mate with the mating connector **2** in the second direction.

(46) As shown in FIGS. 1 to 10, in the illustrated embodiment, each of the safety cover **100** and the end cover **15** has a front end and a rear end which are opposite to each other in the first direction. When the safety cover **100** is moved from the safe position to the initial position, the front end face of the end cover **15** rests against the rear end face of the safety cover **100** to limit the safety cover **100** at the initial position.

(47) As shown in FIGS. 1 to 10, in the illustrated embodiment, the housing **10** includes at least two housing units and a connection part **10a**. The connection part **10a** is connected between two adjacent housing units of the housing **10**. A mounting hole is formed on the connection part **10a**, and the mechanical connection member **12** is installed in the mounting hole of the connection part **10a**.

(48) As shown in FIGS. 1 to 10, in the illustrated embodiment, connector **1** includes at least two terminals **13**, at least two connection assemblies **11**, at least two cables **14** and at least two end covers **15**. At least two terminals **13** are respectively arranged in at least two housing units for electrical connection with at least two mating terminals **21**. At least two connection assemblies **11** are respectively installed in at least two housing units for fastening at least two terminals **13** to at least two mating terminals **21**. At least two cables **14** are respectively inserted into at least two housing units and electrically connected with at least two terminals **13**. At least two end covers **15** are respectively sleeved on one ends of at least two housing units for fixing at least two cables **14** to at least two housing units.

(49) However, the present invention is not limited to the illustrated embodiment. For example, the connector **1** can include: one terminal **13**, one connection assembly **11**, one cable **14** and one end cover **15**.

(50) As shown in FIGS. 1 to 10, in the illustrated embodiment, the safety cover **100** is an integral molded part. For example, the safety cover **100** can be integrally molded.

(51) As shown in FIGS. 1 to 12, in the illustrated embodiment, an elastic cantilever **102b** is formed on the safety cover **100**, and a limit protrusion **10d** is formed on the housing **10**. As shown in FIG. 11, when the housing **10** is not mated with the mating housing **20** and the safety cover **100** is in the initial position, the elastic cantilever **102b** is in the position of interference with the limit protrusion **10d** to prevent the safety cover **100** from being moved from the initial position to the safe position.

(52) As shown in FIGS. 1 to 12, in the illustrated embodiment, when the housing **10** has been mated with the mating housing **20** and the safety cover **100** is in the initial position, the elastic cantilever **102b** is pushed by the unlocking part **20d** on the mating housing **20** to a position that does not interfere with the limit protrusion **10d**, so that the movement of the safety cover **100** will not be hindered by the limit protrusion **10d**.

(53) As shown in FIGS. 1 to 12, in the illustrated embodiment, when the housing **10** has been mated with the mating housing **20** and the safety cover **100** has been moved to the safe position, the free end of the elastic cantilever **102b** rests against the stop part **20b** of the mating housing **20** to limit the safety cover **100** at the safe position.

(54) As shown in FIGS. 1 to 12, in the illustrated embodiment, a threaded hole **210** is formed on the mating terminal **21** or a threaded hole **210** is formed on the mating connection assembly **22** assembled with the mating terminal **21**, and the connection assembly **11** includes a threaded connection member **110** connected to the threaded hole **210**. A threaded connection hole **220** is formed on the mating housing **20**, and the mechanical connection member **12** of the connector **1** has a threaded connection part **120** connected to the threaded connection hole **220**.

(55) As shown in FIGS. 1 to 12, in the illustrated embodiment, the mating connector **2** also includes an inner housing **200** made of insulation material. The inner housing **200** is installed in the mating housing **20**. In the illustrated embodiment, the housing **10** and the mating housing **20** are conductive housings made of conductive material. The mating housing **20** has a flange portion **20a** extending radially, and the threaded connection hole **220** is formed on the flange portion **20a**.

(56) As shown in FIGS. 1 to 16, in an exemplary embodiment of the present invention, a connector

1 is also disclosed. The connector **1** includes: a housing **10**, a terminal **13**, a connection assembly **11**, a safety cover **100** and a high-voltage interlock **16**. The housing **10** is formed with a slot **1016** and a mounting port **1011**. The terminal **13** is arranged in the housing **10**. The connection assembly **11** is installed in the mounting port **1011** of the housing **10** and is suitable for fastening the terminal **13** to the mating terminal **21** of the mating connector **2**. The safety cover **100** is movably mounted on the housing **10** and can be moved between the initial position where the connection assembly **11** is not covered by the safety cover **100** and the safety position where the connection assembly **11** is covered by the safety cover **100**. The high voltage interlock **16** is movably installed in the slot **1016** and can be moved between the connected position where the high voltage interlock **16** is engaged with the mating high voltage interlock **26** of the mating connector **2** and the disconnected position where the high voltage interlock **16** is separated from the mating high voltage interlock **26**.

(57) As shown in FIGS. **1** to **16**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when the connection assembly **11** is not fastened, the connection assembly **11** is in the position of interference with the safety cover **100** to prevent the safety cover **100** from being moved from the initial position to the safe position.

(58) As shown in FIGS. **1-16** and FIGS. **19-21**, in the illustrated embodiment, the safety cover **100** and the high voltage interlock **16** are connected in a sliding way to make them linked according to the following relationship: when the safety cover **100** is moved to the initial position, the high voltage interlock **16** is moved to the disconnected position; and when the safety cover **100** is moved to the safety position, the high-voltage interlock **16** is moved to the connected position.

(59) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the high-voltage interlock **16** includes an insulator **161** and two detection terminals **163**. The two detection terminals **163** are fixed to the insulator **161** for electrical connection to positive and negative electrodes of a high-voltage interlock circuit (not shown). When the high-voltage interlock **16** is moved to the connected position, the two detection terminals **163** of the high-voltage interlock **16** are electrically connected with the mating detection terminal **263** in the mating high-voltage interlock **26** to turn on the high-voltage interlock circuit. When the high-voltage interlock **16** is moved to the disconnected position, the two detection terminals **163** are separated from the mating detection terminal **263** to cut off the high-voltage interlock circuit.

(60) As shown in FIGS. **1** to **21**, in the illustrated embodiment, when the connector **1** is mated with the mating connector **2**, and when the connection assembly **11** is not fastened, the connection assembly **11** is in the position of interference with the safety cover **100**, so that the safety cover **100** cannot be moved from the initial position to the safe position. Therefore, when the connection assembly **11** is not fastened, the high-voltage interlock **16** is always in the disconnected position. Only when the connection assembly **11** is fastened, the high-voltage interlock **16** can be moved to the connected position.

(61) As shown in FIGS. **1** to **21**, in the illustrated embodiment, during mating the connector **1** with the mating connector **2**, the high-voltage interlock circuit is turned on only after the high-voltage terminals (terminal **13** and terminal **21**) are electrically connected; During the separation of connector **1** and mating connector **2**, the high-voltage interlock circuit is cut off before the high-voltage terminals (terminal **13** and terminal **21**) are separated. In this way, the high-voltage terminals (terminal **13** and terminal **21**) can be avoided from being hot plugged and unplugged, and the use safety of the connector is improved.

(62) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the safety cover **100** is configured to be moved between the initial position and the safety position in the first direction (the horizontal direction in the figures). The high-voltage interlock **16** is configured to be moved between the connected position and the disconnected position in the second direction (the vertical direction in the figures) perpendicular to the first direction.

(63) As shown in FIGS. **1** to **21**, in the illustrated embodiment, a guide slot **10b** extending in the first direction is formed on one of the inner wall surface of the safety cover **100** and the outer wall

surface of the housing **10**, and a guide rib **102a** extending in the first direction is formed on the other. The guide slot **10b** is slidably mated with the guide rib **102a** to guide the safety cover **100** to move between the initial position and the safety position in the first direction relative to the housing **10**.

(64) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the slot **1016** on the housing **10** extends in the second direction, and the insulator **160** of the high-voltage interlock **16** includes the first part **161** extending in the second direction. The first part **161** of the insulator **160** is slidably mated with the slot **1016** of the housing **10** to guide the high-voltage interlock **16** to move between the connected position and the disconnected position in the second direction relative to the housing **10**.

(65) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the insulator **160** of the high-voltage interlock **16** also includes a second part **162** connected with the first part **161**, and the second part **162** is located outside the housing **10**. The detection terminal **163** is fixed to the first part **161** of the insulator **160**, and the second part **162** of the insulator **160** is slidably connected with the safety cover **100**.

(66) As shown in FIGS. **1** to **21**, in the illustrated embodiment, a pair of support walls **103** is formed on the inner wall surface of the top wall **101** of the safety cover **100**, and a part of the second part **162** of the insulator **160** is arranged in a gap between the pair of support walls **103**. A chute **1620** is formed on each side of the second part **162** of the insulator **160**. The pair of support walls **103** are respectively formed with convex parts **103a** which are slidably connected with the chutes **1620**. The chute **1620** includes a main part **1623** extending a predetermined length in a direction inclined to the first direction to convert the movement of the safety cover **100** in the first direction (the horizontal direction in the figure) into the movement of the high-voltage interlock **16** in the second direction (the vertical direction in the figure).

(67) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the chute **1620** also includes a first end part **1621** and a second end part **1622** which are respectively connected with the two ends of the main part **1623** of the chute **1620**, and the first end part **1621** and the second end part **1622** extend in the first direction. When the safety cover **100** is moved to the initial position, the convex part **103a** is moved to the first end part **1621** of the chute **1620**. When the safety cover **100** is moved to the safety position, the convex part **103a** is moved to the second end part **1622** of the chute **1620**.

(68) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the high-voltage interlock **16** also includes a sealing ring **164**. The seal ring **164** is sleeved on the first part **161** of the insulator **160** to seal the gap between the first part **161** of the insulator **160** and the slot **1016** of the housing **10**.

(69) As shown in FIGS. **1** to **21**, in the illustrated embodiment, the mating connector **2** includes an inner housing **200** made of insulation material, and the mating high-voltage interlock **26** is fixedly installed in the inner housing **200**. The mating high-voltage interlock **26** includes an insulator **260** and a mating detection terminal **263** arranged in the insulator **260**. When the high-voltage interlock **16** is mated with the mating high-voltage interlock **26**, the mating detection terminal **263** is bridged between the two detection terminals **163** to connect the two detection terminals **163**.

(70) As shown in FIGS. **1** to **21**, in an exemplary embodiment of the present invention, a connector assembly is also disclosed, which includes a connector **1** and a mating connector **2** suitable for mating with connector **1**. The mating connector **2** includes a mating housing **20**, a mating terminal **21** and a mating high-voltage interlock **26**. The mating housing **20** is suitable for mating with the housing **10** of the connector **1**. The mating terminal **21** is arranged in the mating housing **20** for electrical connection with the terminal **13** of the connector **1**. The mating high-voltage interlock **26** is arranged in the mating housing **20** for mating with the high-voltage interlock **16** of the connector **1**.

(71) Please note that the present invention is not limited to the illustrated embodiments. The technical solutions of the present invention apply, for example, the following two variations:

Example of the First Variation

(72) Refer to FIGS. 1-12 above. In the first variation, a connector **1** is disclosed, which includes a housing **10**, a mechanical connection member **12** and a safety cover **100**. The safety cover **100** is slidably mounted on the housing **10**. The mechanical connection member **12** is mounted on the housing **10** and is suitable for threaded connection with the mating housing **20** of the mating connector **2** to fasten the housing **10** to the mating housing **20**. The mechanical connection member **12** has a second operation part **121** exposed from the housing **10**, enabling the mechanical connection member **12** to be fastened or loosened by operating the second operation part **121**.

(73) Refer to FIGS. 1-12 above, during the mating of connector **1** and mating connector **2**, when the mechanical connection member **12** is not fastened, the second operation part **121** is in the position of interference with the safety cover **100** to prevent the safety cover **100** from being moved from the initial position to the safe position.

(74) Refer to the above FIGS. 1-12. When the mechanical connection member **12** has been fastened during the mating of connector **1** and mating connector **2**, the second operation part **121** is in a position that does not interfere with the safety cover **100** to allow the safety cover **100** to be moved from the initial position to the safe position;

(75) Refer to the above FIGS. 1-12, when the safety cover **100** is in the initial position, the second operation part **121** is not covered by the safety cover **100** to allow the mechanical connection member **12** to be fastened or loosened by operating the second operation part **121**;

(76) Refer to FIGS. 1-12 above, when the safety cover **100** is in the safe position, the second operation part **121** is covered by the safety cover **100** to prevent the mechanical connection member **12** from being loosened by operating the second operation part **121**.

Example of the Second Variation

(77) Refer to FIGS. 1-12 above, in the second variation, a connector **1** is disclosed, which includes: a housing **10**, a terminal **13**, a connection assembly **11** and a safety cover **100**. The terminal **13** is arranged in the housing **10**. The connection assembly **11** is installed in the housing **10** and is suitable for fastening the terminal **13** to a mating terminal **21** of a mating connector **2**. The safety cover **100** is slidably mounted on the housing **10**. The connection assembly **11** has a first operation part **111** exposed from the housing **10**, which enables the connection assembly **11** to be fastened or loosened by operating the first operation part **111**.

(78) Refer to FIGS. 1-12 above, during the mating of connector **1** and mating connector **2**, when the connection assembly **11** is not fastened, the first operation part **111** is in the position of interference with the safety cover **100** to prevent the safety cover **100** from being moved from the initial position to the safe position.

(79) Refer to FIGS. 1-12 above, during the mating of connector **1** and mating connector **2**, when the connection assembly **11** has been fastened, the first operation part **111** is in a position that does not interfere with the safety cover **100** to allow the safety cover **100** to be moved from the initial position to the safe position.

(80) Refer to the above FIGS. 1-12, when the safety cover **100** is in the initial position, the first operation part **111** is not covered by the safety cover **100** to allow the connection assembly **11** to be fastened or loosened by operating the first operation part **111**.

(81) Refer to FIGS. 1-12 above. When the safety cover **100** is in the safe position, the first operation part **111** is covered by the safety cover **100** to prevent the connection assembly **11** from being loosened by operating the first operation part **111**.

(82) In the above-mentioned exemplary embodiments according to the present invention, the safety cover can not only reliably prevent the fastened connection assembly from being accidentally loosened, but also can conveniently judge whether the connection assembly have been fastened, thus improving the use safety of the connector.

(83) In addition, in the aforementioned exemplary embodiments according to the present invention, the high-voltage interlock is linked with the safety cover, so it can be ensured that the high-voltage

interlock is always in a state of separation from the mating high-voltage interlock before the terminal of the connector is electrically connected to the mating terminal of the mating connector, thus improving the use safety of the connector.

(84) It should be appreciated for those skilled in this art that the above embodiments are intended to be illustrative, and not restrictive. For example, many modifications may be made to the above embodiments by those skilled in this art, and various features described in different embodiments may be freely combined with each other without conflicting in configuration or principle.

(85) Although several exemplary embodiments have been shown and described, it would be appreciated by those skilled in the art that various changes or modifications may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.

(86) As used herein, an element recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to “one embodiment” of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “comprising” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property.

Claims

1. A connector, comprising: a housing having a mounting port; a connection assembly installed in the mounting port of the housing and adapted to fasten a terminal of the connector to a mating terminal of a mating connector, the connection assembly has a first operation part exposed from the housing and the connection assembly is capable of being fastened or loosened by operating the first operation part; and a safety cover slidably mounted on the housing, the first operation part is in a position of interference with the safety cover to prevent the safety cover from being moved to a safety position when the connection assembly is not fastened.

2. The connector of claim 1, wherein the first operation part is in a position that does not interfere with the safety cover to allow the safety cover to be moved between an initial position and the safety position when the connection assembly has been fastened.

3. The connector of claim 2, wherein the safety cover has a top wall with an operation hole and a pair of side walls connected to the top wall, the first operation part is located outside the safety cover when the safety cover is in the initial position.

4. The connector of claim 2, wherein an elastic cantilever is formed on the safety cover and a limit protrusion is formed on the housing, the elastic cantilever is in a position of interference with the limit protrusion to prevent the safety cover from being moved from the initial position to the safety position when the housing is not mated with the mating housing and the safety cover is in the initial position.

5. The connector of claim 4, wherein the elastic cantilever is pushed by an unlocking part on the mating housing to a position that does not interfere with the limit protrusion when the housing has been mated with the mating housing and the safety cover is in the initial position, so that movement of the safety cover will not be hindered by the limit protrusion.

6. The connector of claim 4, wherein a free end of the elastic cantilever rests against a stop part on the mating housing to limit the safety cover at the safety position when the housing has been mated with the mating housing and the safety cover has been moved to the safety position.

7. The connector of claim 1, further comprising a mechanical connection member installed on the housing and suitable for threaded connection with a mating housing of the mating connector to fasten the housing to the mating housing, the mechanical connection member has a second operation part exposed from the housing and the mechanical connection member is capable of

being fastened or loosened by operating the second operation part, the second operation part is in a position of interference with the safety cover to prevent the safety cover from being moved from the initial position to the safety position when the mechanical connection member is not fastened, the second operation part is not covered by the safety cover to allow the mechanical connection member to be fastened or loosened by operating the second operation part when the safety cover is in the initial position, the second operation part is covered by the safety cover to prevent the mechanical connection member from being loosened by operating the second operation part when the safety cover is in the safety position.

8. The connector of claim 7, wherein the first operation part and the second operation part are in a position that does not interfere with the safety cover to allow the safety cover to be moved from the initial position to the safety position when the connection assembly and the mechanical connection member are fastened.

9. The connector of claim 7, wherein an operation hole is formed on the safety cover, the second operation part is exposed from the safety cover through the operation hole when the safety cover is in the initial position.

10. The connector of claim 9, wherein a limit part is formed on the inner wall surface of a top wall of the safety cover, the limit part rests against the first operation part to limit the safety cover at the safety position when the safety cover is moved from the initial position to the safety position.

11. The connector of claim 1, wherein a guide slot extending in a first direction is formed on one of an outer wall surface of the housing and an inner wall surface of the safety cover, and a guide rib extending in the first direction is formed on the other of the outer wall surface of the housing and the inner wall surface of the safety cover, the guide rib is buckled into the guide slot and is movable along the guide slot in the first direction to guide the safety cover to be moved between the initial position and the safety position relative to the housing.

12. The connector of claim 11, wherein the guide slot is formed on the outer wall surface of the housing and the guide rib is formed on the inner wall surface of a side wall of the safety cover, the guide slot has a guide surface extending in the first direction and the guide rib slides in contact with the guide surface to guide the guide rib to move in the first direction.

13. The connector of claim 1, further comprising a cable having an end inserted into the housing in a first direction and electrically connected with the terminal, and an end cover sleeved on an end of the housing and fixing the cable to the housing.

14. The connector of claim 13, wherein a cable port allowing the cable to be inserted in the first direction is formed on the end of the housing, an installation port is formed on a top of another end of the housing to allow the connection assembly to be inserted in a second direction perpendicular to the first direction, a mating port is formed on a bottom of the another end of the housing to allow the connector to mate with the mating connector in the second direction.

15. The connector of claim 13, wherein each of the safety cover and the end cover has a front end and a rear end which are opposite to each other in the first direction, a front end face of the end cover rests against a rear end face of the safety cover to limit the safety cover at the initial position when the safety cover is moved from the safety position to the initial position.

16. The connector of claim 13, wherein the housing has at least two housing units and a connection part connected between the housing units, a mounting hole is formed on the connection part and a mechanical connection member is installed in the mounting hole on the connection part, the terminal is one of a pair of terminals arranged respectively in the housing units and electrically connecting with a pair of mating terminals, the connection assembly is one of a pair of connection assemblies installed in the housing units and connecting the terminals to the mating terminals, the cable is one of pair of cables inserted into the housing units and electrically connected with the terminals, and the end cover is one of a pair of end covers sleeved on a pair of ends of the housing units to fix the cables to the housing units.

17. The connector of claim 1, wherein the safety cover is an integral molded part.

18. The connector of claim 1, wherein the terminal is arranged in the housing and electrically connected with the mating terminal.

19. A connector assembly, comprising: the connector according to claim 1; and a mating connector matable with the connector, the mating connector includes a mating housing matable with the housing of the connector and a mating terminal disposed in the mating housing for electrical connection with the terminal of the connector.

20. The connector assembly according to claim 19, wherein the mating connector has a mating connection assembly formed with a threaded hole and the connection assembly has a threaded connection member connected into the threaded hole, a threaded connection hole is formed on the mating housing and the mechanical connection member of the connector has a threaded connection part connected into the threaded connection hole.

21. A connector, comprising: a housing; a mechanical connection member installed on the housing and suitable for threaded connection with a mating housing of a mating connector to fasten the housing to the mating housing; and a safety cover slidably mounted on the housing, the mechanical connection member has an operation part exposed from the housing so that the mechanical connection member is capable of being fastened or loosened by operating the operation part, the operation part is in a position of interference with the safety cover to prevent the safety cover from being moved from an initial position to a safety position when the mechanical connection member is not fastened, the operation part is in a position that does not interfere with the safety cover to allow the safety cover to be moved from the initial position to the safety position when the mechanical connection member has been fastened, the operation part is not covered by the safety cover to allow the mechanical connection member to be fastened or loosened by operating the operation part when the safety cover is in the initial position, the operation part is covered by the safety cover to prevent the mechanical connection member from being loosened by operating the operation part when the safety cover is in the safety position.
